

Geodesica Handout

Allyn Rohrbach, Erich Baur

September 14, 2026

Contents

1	Introduction	3
1.1	Definitions	3
1.2	Embeddings, Topology and Curvature	3
1.2.1	Gaussian Curvature	4
1.2.2	Surfaces of constant curvature	4
2	Tessellations of the euclidean plane	5
2.1	Uniform Tilings	5
2.2	Tilings of the plane	5
2.3	Regular Tesselations	5
2.4	Semi- and Quasi-Regular Tessellations	5
2.5	Quasi-Regular Tessellations	5
2.6	Triangle Tessellations	5
2.7	Euler Characteristic	6
3	Tessellations of the sphere	7
3.1	Geometry on the sphere	7
3.1.1	Great Circles	7
3.1.2	Spherical Lunes	7
3.2	Constructing regular tesselations	9
3.2.1	Hosohedron	9
3.2.2	Triangle Tilings	9
3.2.3	Regular and Uniform Tesselations	9
3.3	Excursion Polyhedra	9
4	Stereographic Projection	10
4.1	Polar Coordinates	12
4.2	Distance Metric	12
4.3	Area of a circle	12
5	Standart hyperbolic plane	13
5.1	Hyperbolic Models	13
5.2	Distance Metric	13
5.3	Transformations	13

1 Introduction

A *tessellation* (or Tiling) is the process of covering a surface with geometrical shapes so that there are no gaps or overlaps. The shapes used in a tiling are called *tiles*. A tiling can be made up of one unique or several different tiles. The possibilities of tessellating a surface seem virtually limitless. In the following we are only discussing polygonal edge-to-edge tilings, where tiles are restricted to be polygons which must align with each other on their edges such that they share common vertices.

In this series we take a look a closer look at uniform tilings of surfaces with different geometry. Our goal is to understand the geometric properties of the (flat) euclidean plane, the sphere, as well as the hyperbolic plane.

1.1 Definitions

- ☐ **Regular polygon.** A polygon that is equiangular and equilateral.
- ☐ **Edge transitive.** All edges separate the same type of face.
- ☐ **Face transitive.** All faces are the same (shape)
- ☐ **Vertex transitive.** All vertices are identical (in terms of what faces meet there)
- ☐ **Tiling.** a covering of the surface without gaps or overlaps with polygons
 - ☐ we're only considering edge-to-edge tilings (where polygons edges align to each other) there is another family of tilings that are not edge-to-edge
- ☐ **Uniform tiling.** A tiling of regular polygons that is vertex transitive
- ☐ **Quasi-regular tiling.** A uniform tiling that is edge-transitive
- ☐ **Regular tiling:** A uniform tiling that is edge- and face-transitive
- ☐ **Semi-regular tiling:** A uniform tiling that is neither face- nor edge-transitive
- ☐ **Digon.** A polygon with only two vertices.

https://en.wikipedia.org/wiki/Quasiregular_polyhedron \ https://en.wikipedia.org/wiki/Uniform_tiling

1.2 Embeddings, Topology and Curvature

A surface S is said to be two dimensional if there exists a injective function from $S \rightarrow \mathbb{R}^2$. Meaning we can associate every element in S with a coordinate in $\mathbb{R}^2$.

Let us look at the points of a sphere $S \in \mathbb{R}^3$. A sphere is the set of all points in $\mathbb{R}^3$ that are at a distance r from it's center M . Unlike a Ball a sphere does not have any volume, it is if you will, the boundary surface of the ball. Although the sphere "lives" in three dimensions, for a someone living on it the surface itself appears two dimensional.

We can easily proof that the surface of a sphere is two dimensional using polar coordinates. By assigning identifying each point on the sphere with a latitude

and longitude angle we can describe the entire sphere with a two dimensional coordinate system.

We say that the spherical surface is embedded in three dimensional space.

1.2.1 Gaussian Curvature

Curvature of surface Add informal definition

Cylinder Isometric to the a strip on the euclidean plane

1.2.2 Surfaces of constant curvature

A surfaces a generalization of a plane. A surface is two dimensional the surface can be described by a two dimensional coordinate system

https://en.wikipedia.org/wiki/Differential_geometry_of_surfaces#Surfaces_of_constant_curvature

- how can we proof that the sphere has $k = 1$, the standard hyperbolic plane $k = -1$

2 Tessellations of the euclidean plane

2.1 Uniform Tilings

Uniform tilings are a larger

Definition 1 (Vertex transitive) *A tiling is said to be vertex transitive if all of its vertices are indistinguishable from each other*

- ☐ (playful) take some regular polygons, find arrangements to cover the plane without gapping
- ☐ what is a tessellation
- ☐ what makes it regular

2.2 Tilings of the plane

- ☐ definition face, edge, vertex transitivity

2.3 Regular Tessellations

- ☐ platonic tilings
- ☐ proof that there are only three
- ☐ Schläfli Symbol $\{p, q\}$ describes a tiling where q p -gons meet at each vertex

2.4 Semi- and Quasi-Regular Tessellations

- ☐ archimedean tilings
 - ☐ most are semi-regular derived by truncation
 - ☐ 3.6.3.6 is the only quasi-regular archimedean tiling
- ☐ in a later chapter we can derive semi-regular tilings by truncating, and quasi-regular tilings by rectifying regular tilings.

https://en.wikipedia.org/wiki/Euclidean_tilings_by_convex_regular_polygons#Archimedean,_uniform_or_semi-regular_tilings

2.5 Quasi-Regular Tessellations

- ☐ in a later chapter we can derive quasi regular tilings by rectifying

2.6 Triangle Tessellations

- ☐ I would like to dive deeper into triangle tilings since they're easier to deal with
- ☐ they can serve as a basis to construct a variety of uniform tilings through wythoff construction

$$2 \leq p, q, r$$

2.7 Euler Characteristic

The euler characteristic of a cell division is given by [Weeks, pp.171] Where V is the number of vertices, E the number of edges and F the number of faces

$$\chi = V - E + F$$

For polyhedra $\chi = 2$ since they can be embedded on the sphere $\mathbb{S}^2$ [Coxeter, pp. 10]

The gauss-bonnet formula gives the area of different surfaces

$$A_{\text{sphere}} = 2\pi\chi$$

3 Tessellations of the sphere

To begin our journey into non-euclidean geometries we will first look at geometry of the sphere. Understanding spherical geometry will give us helpful insights to understand and interpret hyperbolic geometry.

3.1 Geometry on the sphere

- ☐ Geodesics (what is the shortest path on the sphere)
- ☐ Antipodal
- ☐ Dihedral Angle

3.1.1 Great Circles

- ☐ Todo, definition for the unit sphere S and representation of planes by unit normal vectors $\hat{n}$

A great circle is given by the intersection of the sphere with a plane through it's center. For any two points on the great circle the arc formed between them is the shortest path the sphere.

- ☐ Show that this intersection is actually a circle?

Let A, B be two points on the sphere. The normal vector of a great circle through A and B is given by the cross product $\vec{A} \times \vec{B}$.

- ☐ We proof that arcs of great circles form the shortest path between two points
- ☐ Note that normal vectors on the xy-plane form great circles through the north and south pole of the sphere.
- ☐ The perpendicular line to a great circle through it's center is called it's axis
- ☐ The two point intersection of a great circles axis with the sphere are called it's poles.
- ☐ A great circle splits the sphere into two hemispheres

Let $x \in S$ be the elements of a sphere and $\hat{n}$ the unit normal vector of a great circle. The great circle partitions the sphere into two hemispheres:

- A positive hemisphere $S_+ = \{x \in S : x \cdot \hat{n} > 0\}$,
- a negative hemisphere $S_- = \{x \in S : x \cdot \hat{n} < 0\}$,

3.1.2 Spherical Lunes

- ☐ Special case of a digon
- ☐ Polygon with two vertices (possible in S, degenerate in E)

Let g_1, g_2 be two great circles and n_1, n_2 their respective unit normal vectors. The two cricles intersect at an angle θ given by $\cos(\theta) = \hat{n}_1 \cdot \hat{n}_2$. Let S_{1+}, S_{2+} be the positive hemisphere given by the great circles. We define a spherical lune L to be the wedge shaped intersection of the two hemispheres $L = S_{1+} \cap S_{2+}$.

The area of a lune is given by it's angle $A(L_\theta) = 2\theta r^2$. A proof is left as an exercise to the reader :) This fact will proof useful later.

Note that the intersection $L' = S_{1-} \cap S_{2-}$ is a congruent antipodal lune to L . Meaning $A(L_\theta) = A(L'_\theta)$.

□ Not sure if we should add this. It is rigorous but also obfuscating the point a lot

□ $\bar{L} = \overline{S_{1+}} \cap \overline{S_{2+}}$ lune including boundary

□ $A(L) = A(\bar{L})$ including boundary does not change area

Spherical Triangles Let $A, B, C \in S$ be points on the unit sphere forming the vertices of a spherical triangle. It's edges a, b, c are given by great circles with respective unit normal vectors $\hat{n}_a, \hat{n}_b, \hat{n}_c$ with

$$n_a = C \times B$$

$$n_b = A \times C$$

$$n_c = B \times A$$

It is good to be rigerous at this stage with how we define the great circles that form our triangle. We want $\Delta = S_{a+} \cap S_{b+} \cap S_{c+}$. This will proof useful once we get into algorithms to generate tilings. Note this also has to be considered in euclidean geometry when we represent lines in their normal form.

$$\alpha = n_b \cdot n_c$$

$$\beta = n_c \cdot n_b$$

$$\gamma = n_a \cdot n_b$$

The angle sum of a spherical triangle is greater than π . There is an elegant proof to show this.

The unit normal vectors are non-colinear and non-planar if and only if A, B, C span a triangle. This is exactly the case when the scalar triple product $[n_a, n_b, n_c] = n_a \cdot (n_b \times n_c) > 0$.

Let us construct the spherical lunes $L_\alpha, L_\beta, L_\gamma$ and their antipodal counterparts $L'_\alpha, L'_\beta, L'_\gamma$. Note that each lune contains either the original triangle Δ or it's antipodal counterpart Δ' . Note further that we can reconstruct the sphere by forming the unions of all lunes $\bigcup \bar{L} = S$.

Summing up the area of all lunes we get back the area of the sphere, however overcounting by four areas of our original triangle. Note our lemma $A(L) = A(L')$ hence $A(L) + A(L') = 2A(L)$. Simplifying we get an equation for the angle sum.

$$\begin{aligned}
2A(L_\alpha) + 2A(L_\beta) + 2A(L_\gamma) &= 4\pi r^2 + 4A(\triangle) \\
&\Downarrow \\
A(L_\alpha) + A(L_\beta) + A(L_\gamma) &= 2\pi r^2 + 2A(\triangle) \\
&\Downarrow \\
2\alpha r^2 + 2\beta r^2 + 2\gamma r^2 &= 2\pi r^2 + 2A(\triangle) \\
&\Downarrow \\
\alpha + \beta + \gamma &= \pi + \frac{A(\triangle)}{r^2}
\end{aligned}$$

This gives us a nice equation for the angle sum in relation to the triangles area. Let us consider the area bounds of a spherical triangle. It is clear that the area of a triangle must be greater than 0 and cannot exceed 2π . Therefore $0 < T < 2\pi$

Further we get a bound on the angle sum $\pi < \alpha + \beta + \gamma < 3\pi$.

This also gives us a formula for the area of a spherical triangle where

$$A(\triangle) = \alpha + \beta + \gamma - \pi$$

3.2 Constructing regular tessellations

3.2.1 Hosohedron

- ☐ Tilings of the sphere by lunes

3.2.2 Triangle Tilings

- ☐ split hosohedron along equator
- ☐ combinations for pqr with angle sum constraint

3.2.3 Regular and Uniform Tessellations

- ☐ Dual graphs on triangle tilings
 - ☐ already getting into kaleidoscope construction through symmetry groups?

3.3 Excursion Polyhedra

- ☐ Tilings of the sphere give us graphs of 3d polyhedra sharing same symmetries and duality
 - ☐ Platonic solids
 - ☐ Prisms

4 Stereographic Projection

Now that we have an understanding of how lines and triangles behave on a sphere

As of now we have studied two-dimensional spherical geometry embedded on the surface of a three-dimensional object, the sphere. This is obviously very natural, yet it begs the question whether we could somehow flatten the surface of the sphere.

We already use projections for our figures, where the sphere is orthographically or perspective projected onto the 2d image. Which our brains are very used to seeing, we are able to reconstruct a 3d interpretation in our minds eye.

The questions is whether there are other projections that might prove more useful in comparing spherical geometry to euclidean and hyperbolic geometry. We want our projection to have useful properties.

- The projection should be bijective in order for us to have one to one correspondance between points in the projection and points in the embedding.
- it should preserve “useful” geometric properties
 - conformal (angle preserving)
 - geodesics should map into ”intuitive” counterpart
- it should come with a metric

Let $\Phi : S \subset \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection from the sphere to the plane and $\Phi^{-1} : \mathbb{R}^2 \rightarrow S \subset \mathbb{R}^3$ the unprojection that from the plane to the spehere. Let $(x, y, z) \in S$ denote points on the sphere and $(X, Y) \in \mathbb{R}^2$ points on the plane

$$\Phi(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z} \right) = (X, Y)$$

$$\Phi^{-1}(X, Y) = \left(\frac{2X}{X^2 + Y^2 + 1}, \frac{2Y}{X^2 + Y^2 + 1}, \frac{X^2 + Y^2 - 1}{X^2 + Y^2 + 1} \right) = (x, y, z)$$

The stereographic projection maps great circles to

- straight lines through the origin, if the great circle intersects the north pole (projection point)
- or circles otherwise

Proof. Let a great circle be given by its unit normal vector $\hat{n} = (n_x, n_y, n_z)$. Points on the great circle are given by

$$xn_x + yn_y + zn_z = 0$$

we substitute x, y, z by unprotecting the corresponding point on the plane (X, Y)

$$\Leftrightarrow n_x \frac{2X}{X^2 + Y^2 + 1} + n_y \frac{2Y}{X^2 + Y^2 + 1} + n_z \frac{X^2 + Y^2 - 1}{X^2 + Y^2 + 1} = 0$$

$$\Leftrightarrow 2Xn_x + 2Yn_y + n_z(X^2 + Y^2 - 1) = 0$$

If $n_z = 0$, in which case the great circle must intersect the north and south pole, the equation degenerates into a linear equation, giving us points on a straight line through the origin.

$$\Leftrightarrow 2Xn_x + 2Yn_y = 0$$

If $n \neq 0$ we divide by n_z

$$\Leftrightarrow 2X \frac{n_x}{n_z} + 2Y \frac{n_y}{n_z} + X^2 + Y^2 - 1 = 0$$

We substitute $a = 2 \frac{n_x}{n_z}$ and $b = 2 \frac{n_y}{n_z}$ for readability and rearrange to

$$\Leftrightarrow (X^2 + aX) + (Y^2 + bY) = 1$$

finally, by completing the square we end up with the equation of a circle

$$\Leftrightarrow \left(X + \frac{a}{2}\right)^2 + \left(Y + \frac{b}{2}\right)^2 = 1 + \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2$$

We can determine the exact properties the projected great circles

In the first case, the line given by the normal vector $\hat{n}' = \begin{pmatrix} n_x \\ n_y \end{pmatrix}$

$$2Xn_x + 2Yn_y = 0$$

$$\Leftrightarrow Xn_x + Yn_y = 0$$

$$\Leftrightarrow \begin{pmatrix} X \\ Y \end{pmatrix} \cdot \begin{pmatrix} n_x \\ n_y \end{pmatrix} = 0$$

In the second case, the circle can be described by its center $C = (c_x, c_y)$ and radius r

The radius is given by

$$r = 1 + \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 = 1 + \frac{n_x^2 + n_y^2}{n_z^2}$$

$$c_x = -\frac{a}{2} = -\frac{n_x}{n_z}$$

$$c_y = -\frac{b}{2} = -\frac{n_y}{n_z}$$

- The circle is smallest with radius $r = 1$ when $\hat{n} = (0, 0, \pm 1)$, i.e. when the great circle lies on the equator
- The radius grows to infinity as $r \xrightarrow{n_z \rightarrow 0} \infty$
- ☐ show how the formula changes if we offset the sphere along the z-axis, while still projecting from it's north pole
- ☐ no isometric mapping
- ☐ to get a better understanding of hyperbolic models in the next chapter, we want to understand the projection S2 onto E2
- ☐ Great circles map to circles or lines
- ☐ Reflection on lines becomes inversion
- ☐ Excursion into mapping of transformations on the sphere
- ☐ Distortions through the projection

4.1 Polar Coordinates

- ☐ r : radius
- ☐ θ : inclination
- ☐ ϕ : azimuth (polar angle)
- ☐ conversion cartesian coordinates

4.2 Distance Metric

The distance between two points on a sphere is given by the minor arc length of a great circle.

- ☐ exercise generalise the formula for a sphere with radius R
- ☐ how we measure distance on the sphere
- ☐ how we measure distance in our model
- ☐ area metric
- ☐ central angle $\Delta\sigma = \mathbf{a} \cdot \mathbf{b}$

4.3 Area of a circle

Let P be a point on the sphere. We define a circle with radius r to be the set of points that are distance r away from P

The central angle is given by $\Delta\sigma = \frac{R}{r}$.

5 Standart hyperbolic plane

- ☐ example
- ☐ infinitelz many tesselations
- ☐ more space

5.1 Hyperbolic Models

Hyperbolic geometry is by itself much harder to grasp.

- ☐ Projection of paraboloid

5.2 Distance Metric

A

5.3 Transformations